

Coupled deployment profiles

The comparison does not isolate precision, graph shape, KV strategy, or compute unit causally

Static-W8

Weights W8 per-tensor K-means palettization

Embedding INT8 per-tensor quantization

KV state FP16 fixed-size

Graph static

AOT h16p; neural-engine

Dynamic-INT4

Weights INT4 block-32 symmetric-with-clipping

Embedding not separately reported in the artifact summary

KV state FP16 growing/dynamic

Graph dynamic

AOT h16p; gpu

Shared: base revision, tokenizer, 4,096-token cap, h16p device class, iOS 27, and benchmark controls